Institute of electrical engineering and electronics

LAB Report # 1

- Submitted by:
  - * Bandani Mohamed Islam
  - * Mahmoudi Mohammed EL Amine
- GROUP: 02
- Field: Computer Engineering (M1)
- Supervisor: DR A. Benzekri

Scanned with CamScanner

LAB WORK #2

PART 01:

1-

we have

$y = \bar{w} Q_1 Q_0$

$D_1 = wQ_0 + w Q_1$

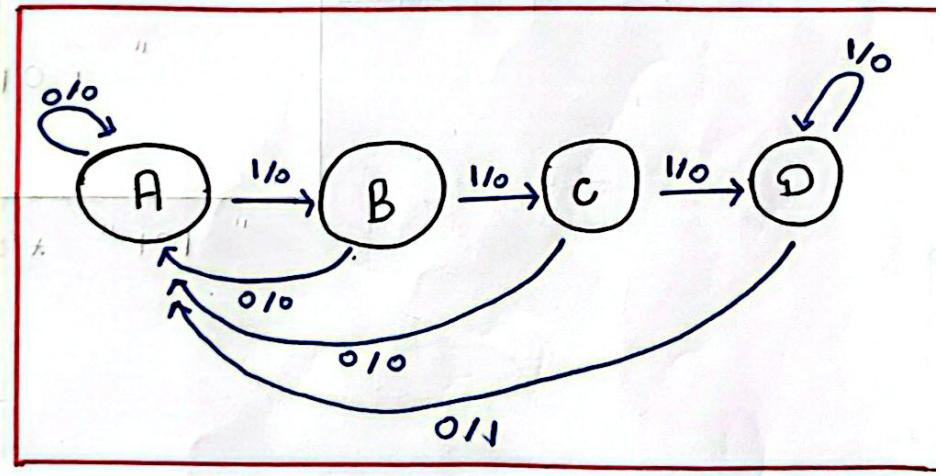
$D_0 = w\bar{Q}_0 + w Q_1$

A = 00
B = 01
C = 10
D = 11

w	$Q_1 Q_0$	$D_1 D_0$	y
0	0 0	0 0	0
0	0 1	0 0	0
0	1 0	0 0	0
0	1 1	0 0	1
1	0 0	0 1	0
1	0 1	1 0	0
1	1 0	1 1	0
1	1 1	1 1	0

$Q_1^+ Q_0^+$

w	P.S	N.S	y
0	A	A	0
0	B	A	0
0	C	A	0
0	D	A	1
1	A	B	0
1	B	C	0
1	C	D	0
1	D	D	0



× "1110" detector ×

The FSM is a mealy model which detects the sequence "1110"

Scanned with CamScanner

2-

$y = Q_1 Q_0$

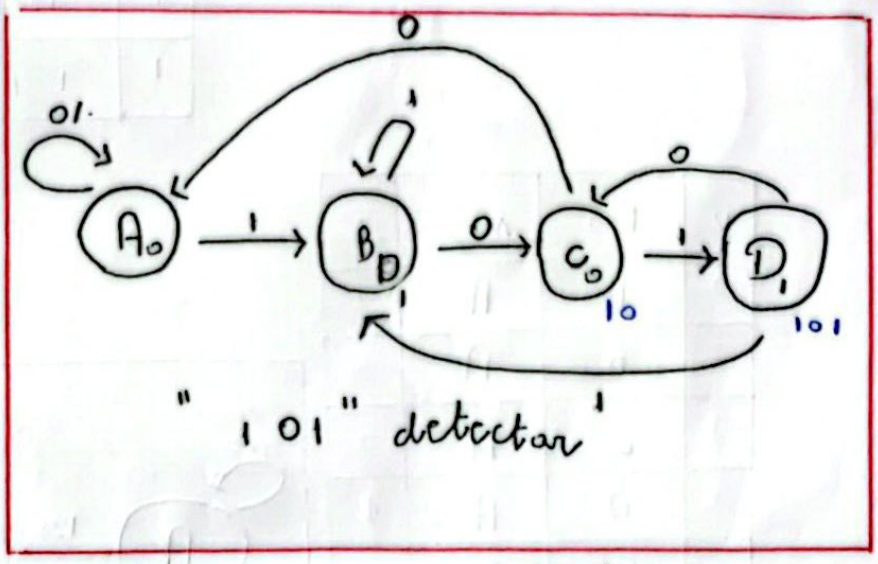
$D_1 = w Q_1 \bar{Q}_0 + \bar{w} Q_0$

$D_0 = w$

w	Q_1 Q_0	D_1 D_0	y
0	0 0	0 0	0
0	0 1	1 0	0
0	1 0	0 0	0
0	1 1	1 0	1
1	0 0	0 1	0
1	0 1	0 1	0
1	1 0	1 1	0
1	1 1	0 1	1

Q_1^+ Q_0^+

w	P.S	N.S	y
0	A	A	0
0	B	C	0
0	C	A	0
0	D	C	1
1	A	B	0
1	B	B	0
1	C	D	0
1	D	B	1

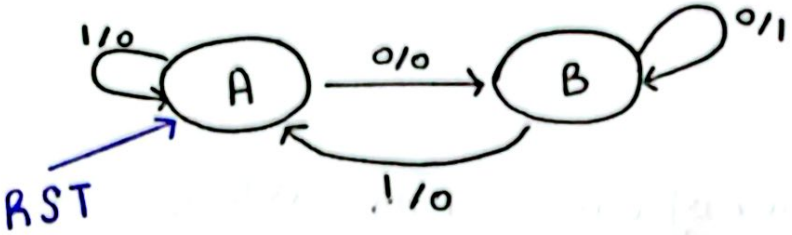


" 101 " * sequence detector *

PART 02:

I- two or more Successive 1ons:

using mealy model FSM:

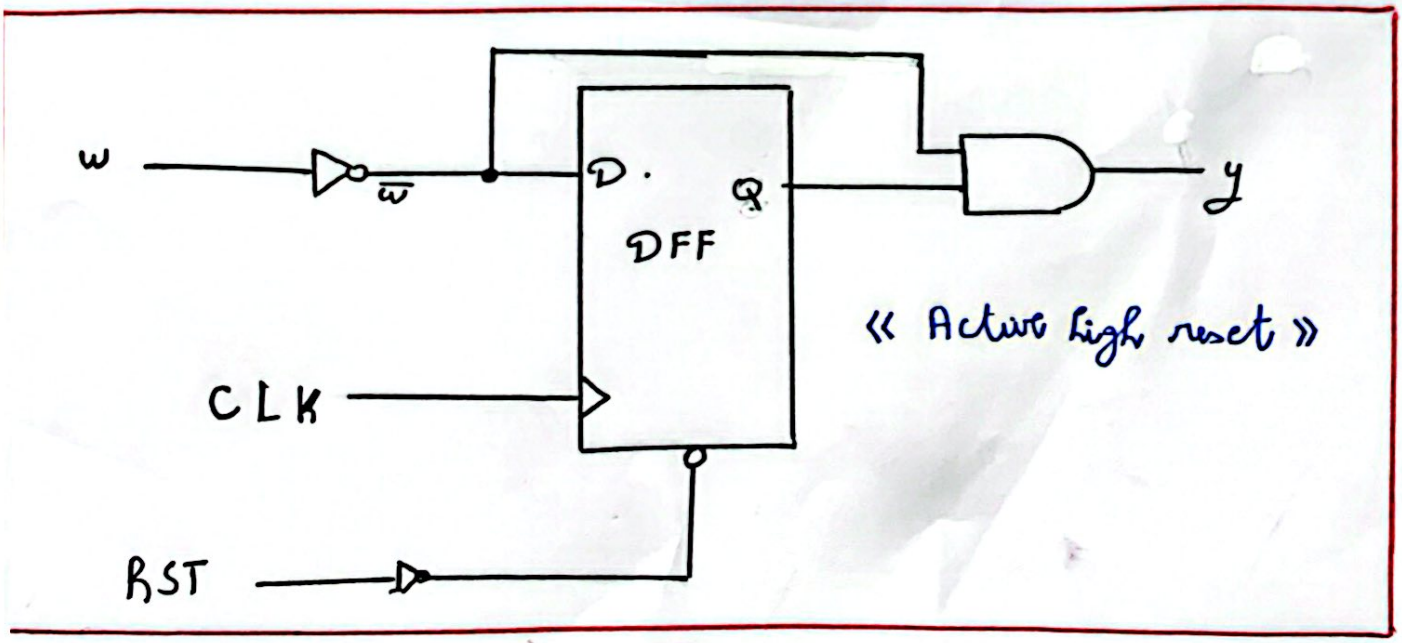


w	P.S	N.S	y
0	A	B	0
0	B	B	1
1	A	A	0
1	B	A	0

⟹

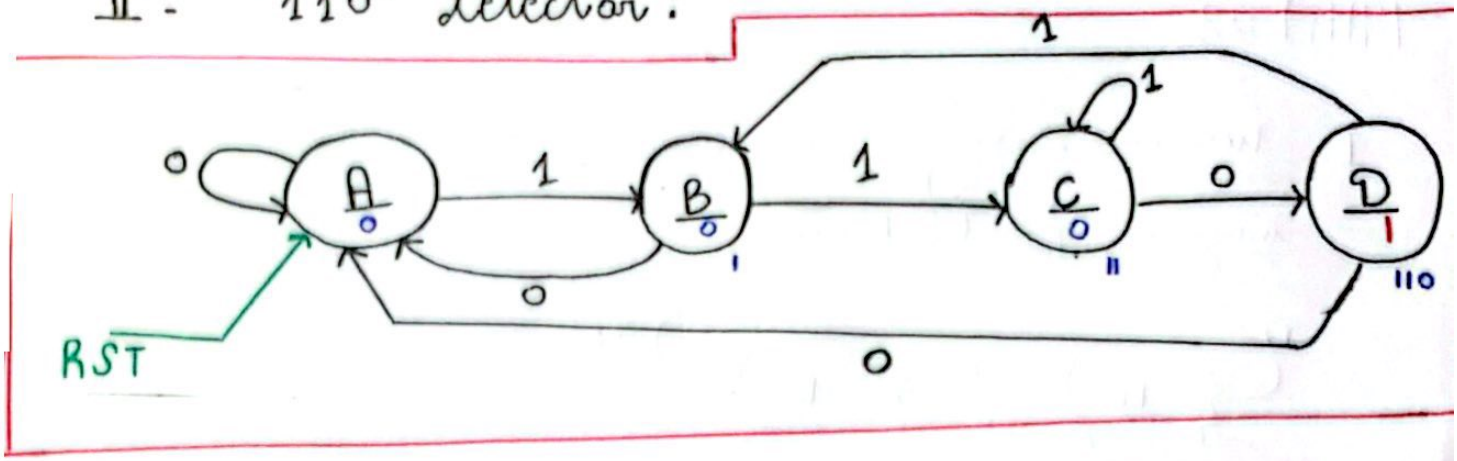
w	Q	D	y
0	0	1	0
0	1	1	1
1	0	0	0
1	1	0	0

Then : $y <= \overline{w}\, Q$ $D <= \overline{w}$



Scanned with CamScanner

II - "110" detector:



- This is the state diagram of a moore FSM, which detects the sequence "110"

w	PS	NS	y
0	A	A	0
0	B	A	0
0	C	D	1
0	D	A	0
1	A	B	0
1	B	C	0
1	C	C	0
1	D	B	0

w	Q_1 Q_0	D_1 D_0	y
0	0 0	0 0	0
0	0 1	0 0	0
0	1 0	1 1	D
D	1 1	0 0	1
1	0 0	D 1	0
1	0 1	1 0	0
1	1 0	1 0	0
1	1 1	0 1	1

$y = Q_1 Q_0$

D_1

w \ Q_1Q_0	00	01	11	10
0	0	0	0	1
1	0	1	0	1

$D_1 = Q_1\bar{Q}_0 + w\bar{Q}_1 Q_0$

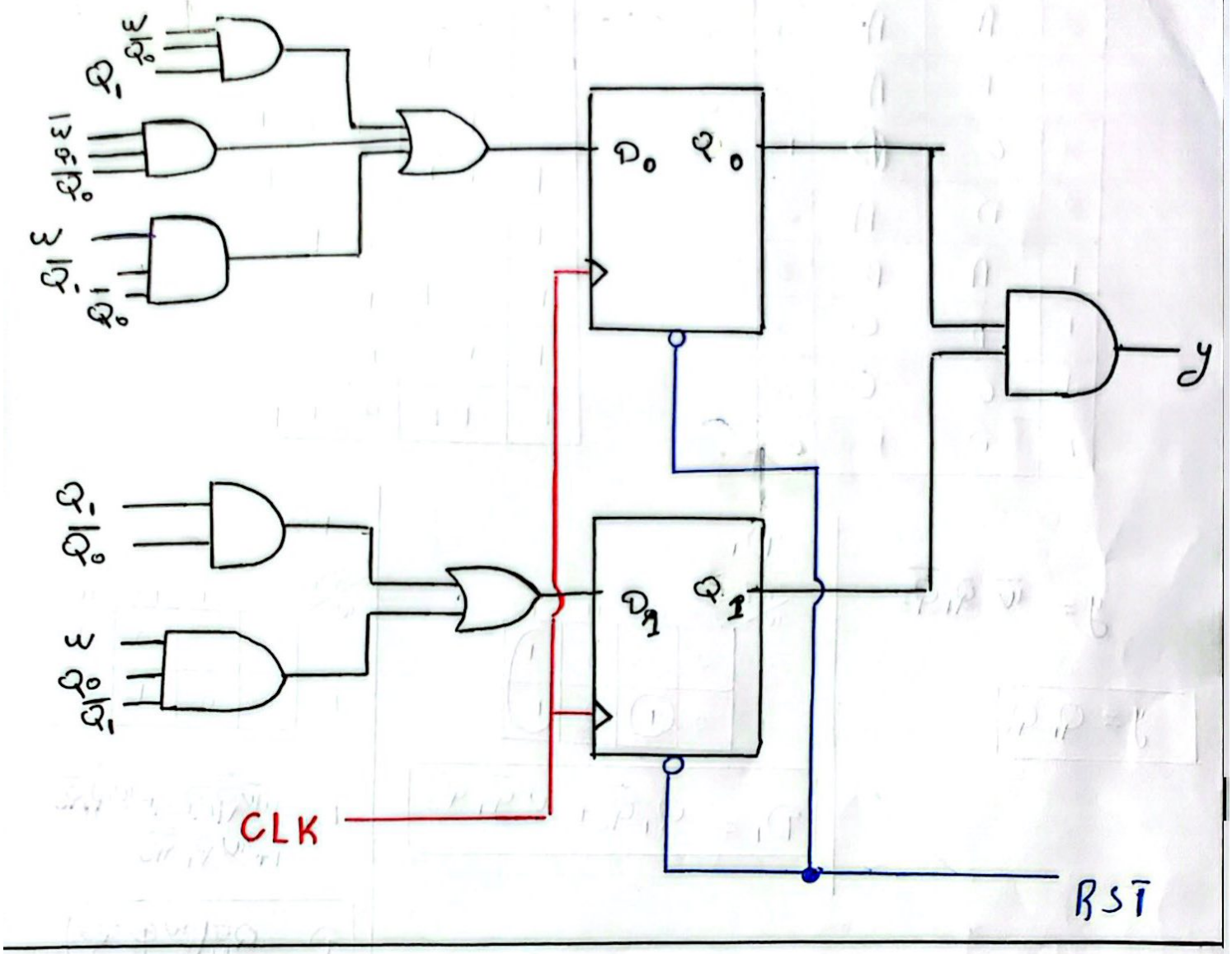
D_0

w \ Q_1Q_0	00	01	10	10
0	0	0	0	1
1	1	0	1	0

$D_0 = w\bar{Q}_1\bar{Q}_0 + \bar{w}Q_1\bar{Q}_0 + wQ_1\bar{Q}_0$

$D_0 = w\bar{Q}_1\bar{Q}_0 + \bar{w}Q_1\bar{Q}_0 + wQ_1\bar{Q}_0$

Scanned with CamScanner

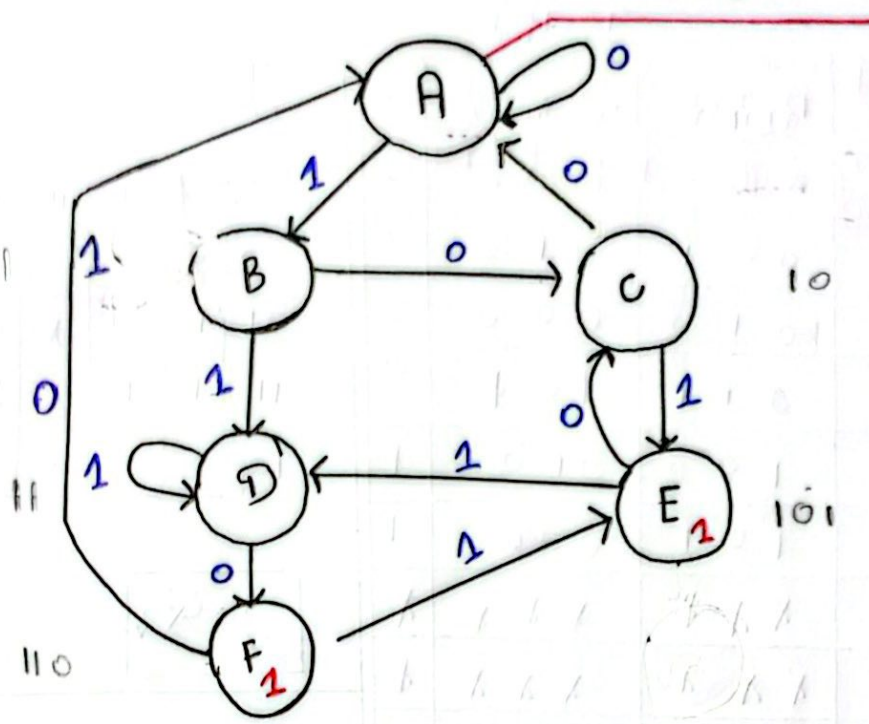


*

- The circuit represents a moore FSM using DFF which detects the sequence "110"

Scanned with CamScanner

III: moore model "110" or "101", overlapping is allowed:


RST
A
B
C
D
E
F
0
1
1
0
0
1
0
1
1
0
1
1
0
1

		$D_2\ D_1\ D_0$	
w	$Q_2 Q_1 Q_0$	$Q_2^+ Q_1^+ Q_0^+$	y
0	0 0 0	0 0 0	0
0	0 0 1	0 1 0	0
0	0 1 0	0 0 0	0
0	0 1 1	1 0 1	0
0	1 0 0	0 1 0	1
0	1 0 1	0 0 0	1
X	X X X	X X X	X
X	X X X	X X X	X
1	0 0 0	0 0 1	0
1	0 0 1	0 1 1	0
1	0 1 0	1 0 0	0
1	0 1 1	0 1 1	0
1	1 0 0	0 1 1	1
1	1 0 1	1 0 1	1
X	X X X	X X X	X
X	X X X	X X X	X

y:

wQ_2 \ Q_1Q_0	00	01	11	10
00	0	0	0	0
01	1	1	X	X
11	1	1	X	X
10	0	0	0	0

$$y = Q_2$$

D_2:

wQ_2 \ Q_1Q_0	00	01	11	10
00	0	0	1	0
01	0	0	X	X
11	0	0	X	X
10	0	0	0	1

$$D_2 = \bar{w} Q_1 Q_0 + w Q_2 Q_0 + w Q_1 \bar{Q}_0$$

D_1:

wQ_2 \ Q_1Q_0	00	01	11	10
00	0	1	0	0
01	1	0	X	X
11	1	0	X	X
10	0	1	1	0

$$D_1 = Q_2 \bar{Q}_0 + w \bar{Q}_2 Q_0 + \bar{w} \bar{Q}_2 \bar{Q}_1 Q_0$$

D_0:

wQ_2 \ Q_1Q_0	00	01	11	10
00	0	0	1	0
01	0	0	X	X
11	1	1	X	X
10	1	1	1	0

$$= w \bar{Q}_1 + w Q_0 + \bar{w} Q_1 Q_0$$

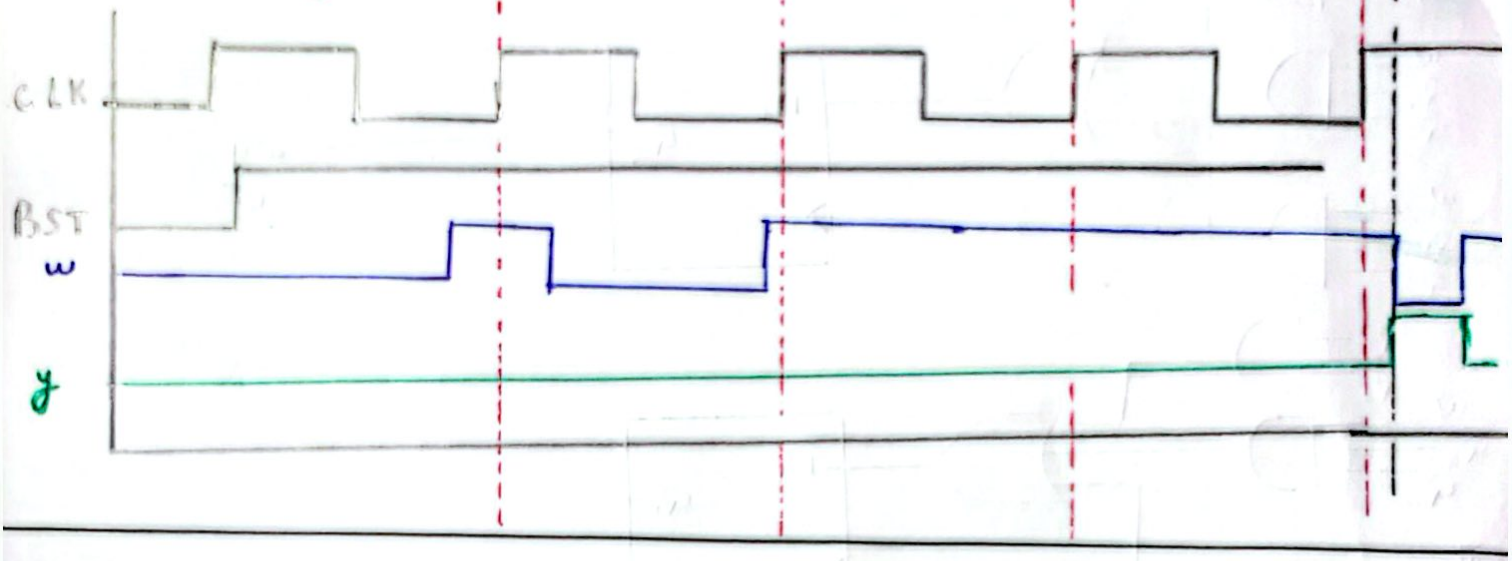
Scanned with CamScanner

D_2
Q_2
y
D_1
Q_1
D_0
Q_0

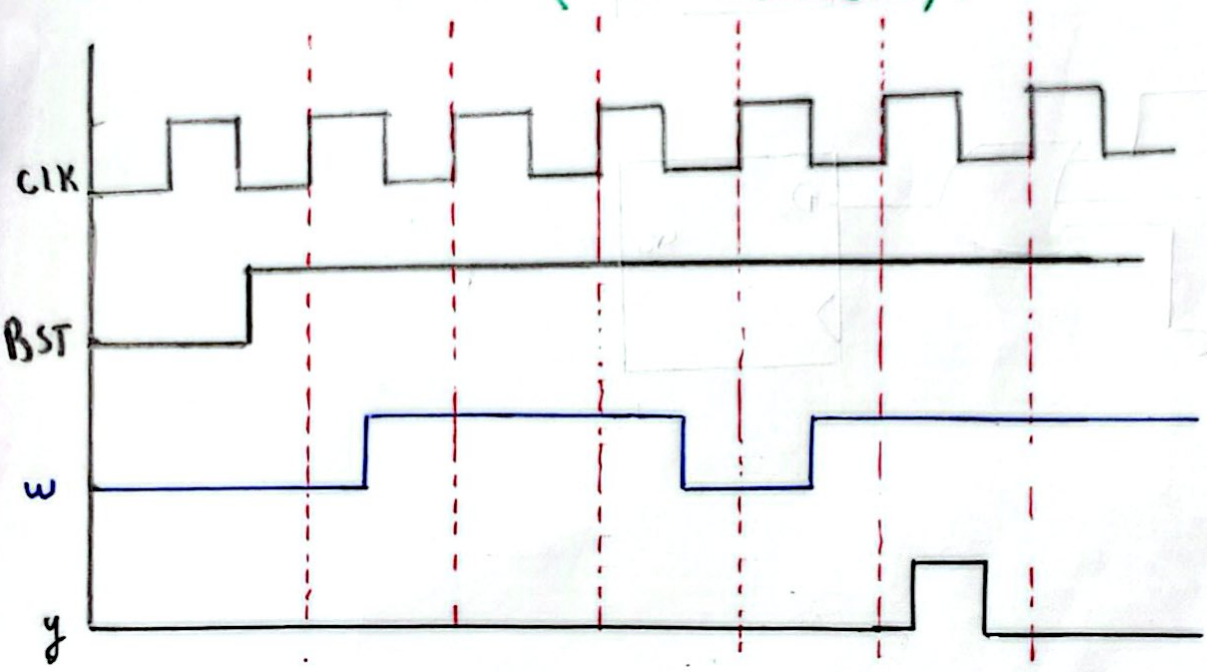
* Functional simulations:

- PART I:

1- mealy model FSM ("1110" detector):



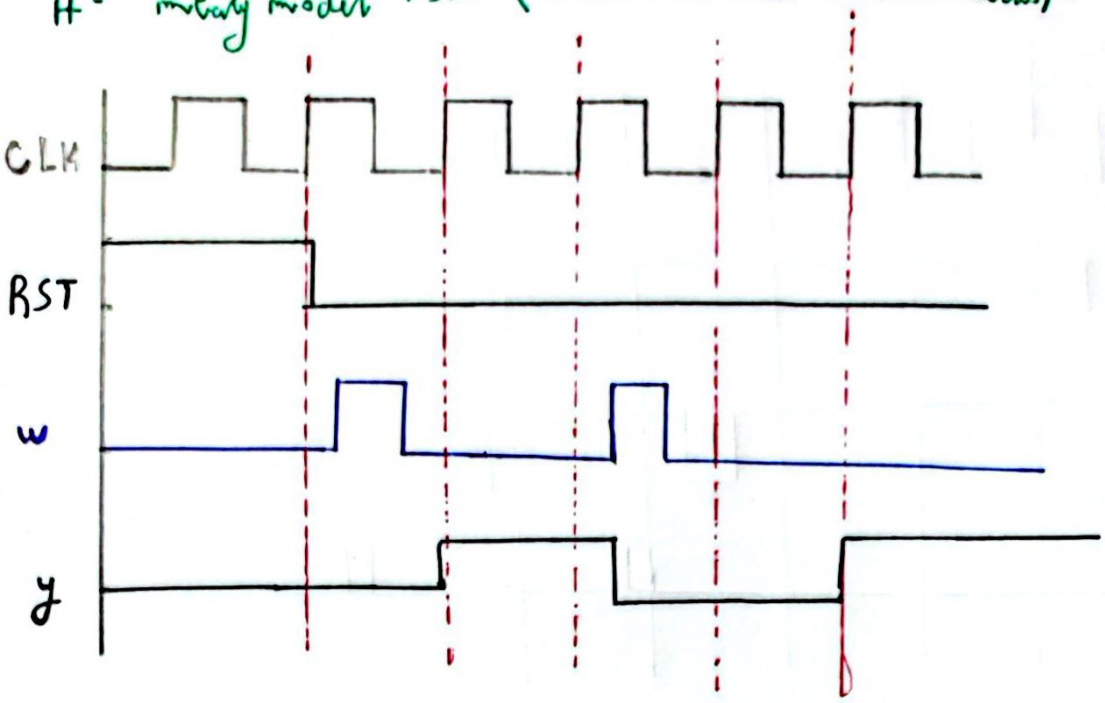
2- moore model FSM ("101" detector):



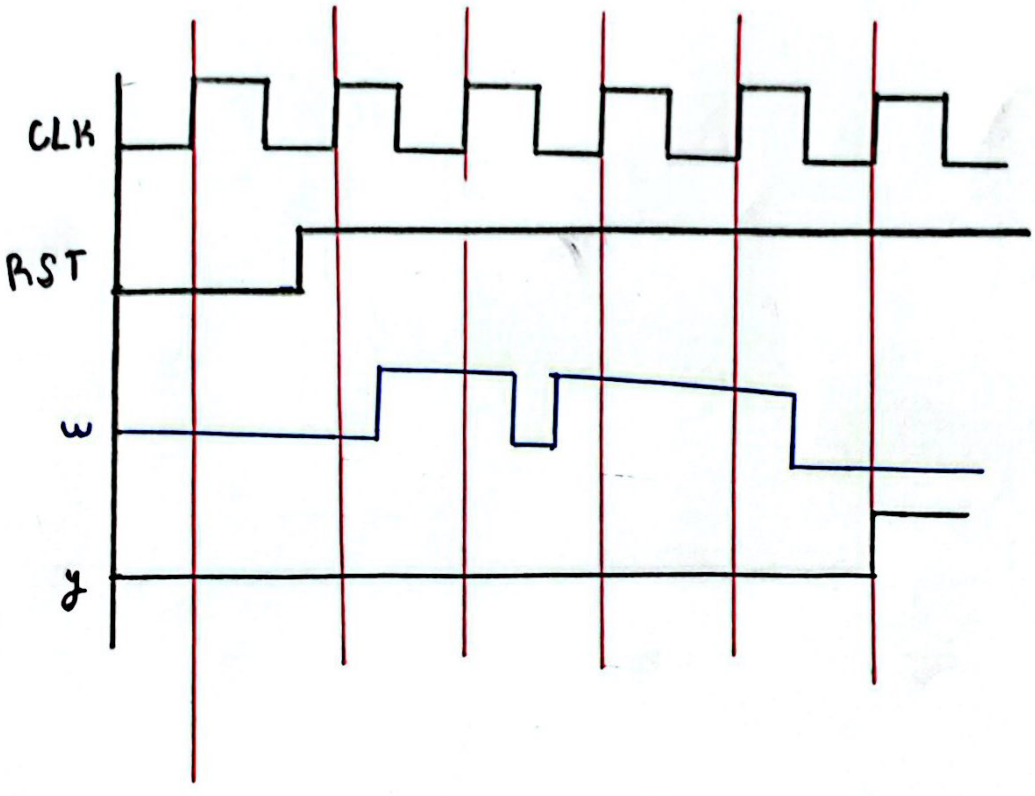
Scanned with CamScanner

- PART II:

A- mealy model FSM (two or more successive Zeros):

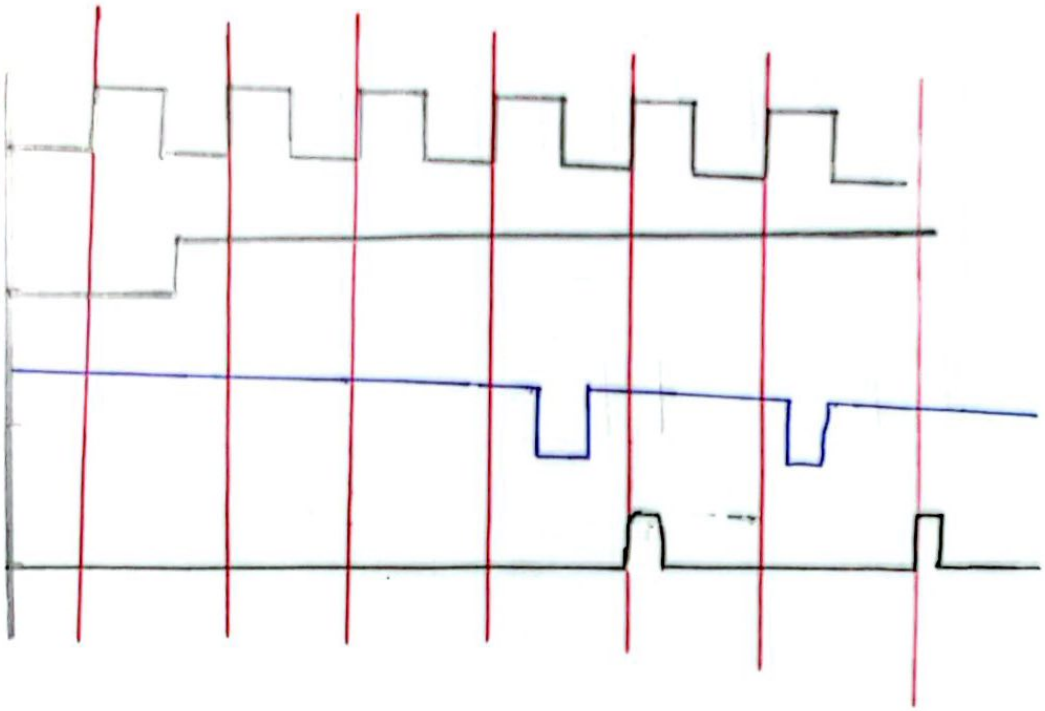


B- moore model FSM ("110" detector")



Scanned with CamScanner

C - moore model FSM ("110" or "101"):



Scanned with CamScanner